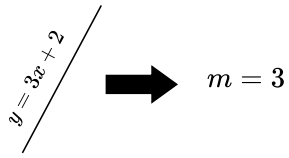
1.30 Differentiation from First Principles

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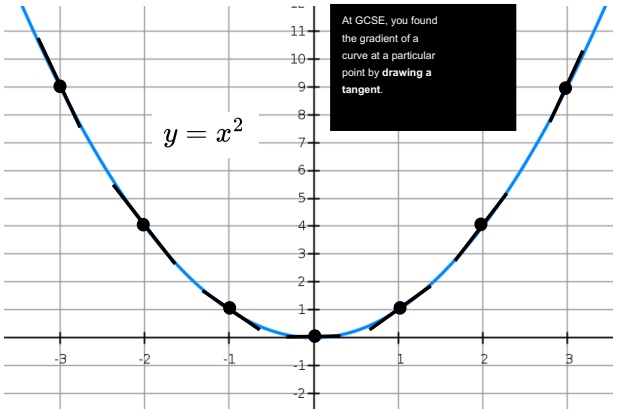
Gradient Function

3

For a straight line, the gradient is **constant**:



However, for a curve **the gradient varies**. We can no longer have a single value for the gradient; **we ideally want an expression in terms of $\mathbf{x}$** that gives us the gradient for any value of x (unsurprisingly known as the **gradient function**).



By looking at the relationship between x and the gradient at that point, can you come up with an expression, in terms of _____ for the gradient?

Gradient Function _____

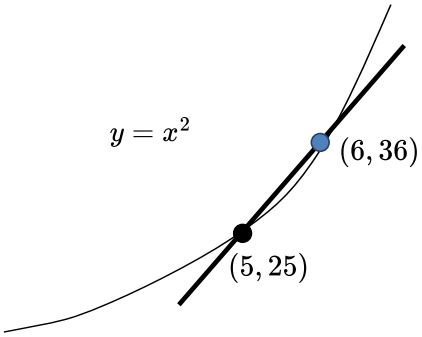
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Finding the Gradient Function

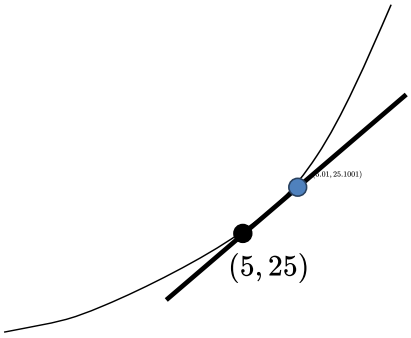
4

The question is then: Is there a method to work out the gradient function without having to draw lots of tangents and hoping that we can spot the rule?

To approximate the gradient on the curve $y = x^2$ when $x = 5$, we could pick a point on the curve just slightly to the right, then find the gradient between the two points:



As the second point gets closer and closer, the gradient becomes a better approximation of the true gradient:



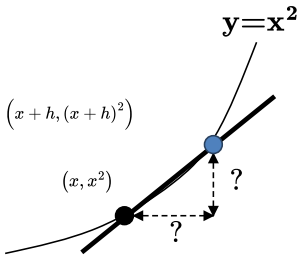
1.30 Differentiation from First Principles

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Finding the Gradient Function

5

This gives us a numerical method to get the gradient **at a particular x**, but doesn't give us the gradient function in general. Let's use exactly the same method, but keep x general, and make the 'small change' (which was previously 0.01) 'h':



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Finding the Gradient Function

6

The gradient function, or derivative, of the curve $y = f(x)$ is written as $f\prime(x)$ or $\frac{dy}{dx}$.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

The gradient function can be used to find the gradient of the curve for any value of x.

We will soon see an easier/quicker way to differentiate expressions like $y = x^3 - x$ without using 'limits'. But this method, known as **differentiating by first principles**, is now in the A Level syllabus, and you could be tested on it!

Notation Note:

Whether we use $\frac{dy}{dx}$ or $f\prime(x)$ for the gradient function depends on whether we use $y =$ or $f(x) =$ to start with:

"Leibniz's notation"

$$y = x^2 \quad \rightarrow \quad \frac{dy}{dx} = 2x$$
$$f(x) = x^2 \quad \rightarrow \quad f'(x) = 2x$$

"Lagrange's notation"

Advanced Notation Note:

Rather than h for the small change in x, the formal notation is δx. So actually:

$$f'(x) = \lim_{\delta x\to 0} \frac{f(x+\delta x)-f(x)}{\delta x}$$

So we in fact have 3 symbols for "change in"!

- Δx: any change in x (as seen in Chp5: $m = \frac{\Delta y}{\Delta x}$)
- δx: a small change in x
- dx: an infinitesimally small change in x

There's in fact a third way to indicate the gradient function, notation used by Newton: (but not used at A Level)

$$y = x^2 \quad \rightarrow \quad \dot{y} = 2x$$

So the estimated gradient using some point close by was $\frac{\delta y}{\delta x}$, but in the 'limit' as $\delta x \to 0$, $\frac{\delta y}{\delta x} \to \frac{dy}{dx}$

1.30 Differentiation from First Principles

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Example

7

The point A with coordinates $(4, 16)$ lies on the curve with equation $y = x^2$.
At point A the curve has gradient g.
a) Show that $g = \lim_{h \to 0} (8 + h)$
b) Deduce the value of g.

Use the “differentiation by first principles” formula.

Function is $f(x) = x^2$

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Test Your Understanding

8

Prove **from first principles** that the derivative of x^4 is $4x^3$.

Support:
“Row 4” of Pascal’s Triangle is:
1 4 6 4 1
You’re welcome.

As $h \to 0$, any term involving a multiplication by h will become 0.

1.30 Differentiation from First Principles

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Exercise 12B – Pearson Pure Y1 page 261-262

9

5 Prove, from first principles, that the derivative of $6x$ is 6.

Reminder:
$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.

$$f'(x) = \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

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Exercise 12B – Pearson Pure Y1 page 261-262

10

6 Prove, from first principles, that the derivative of $4x^2$ is $8x$.

Reminder:
$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.

$$f'(x) = \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

1.30 Differentiation from First Principles

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Exercise 12B – Pearson Pure Y1 page 261-262

11

7 $\mathrm{f}(x) = ax^2$, where a is a constant. Prove, from first principles, that $\mathrm{f}'(x) = 2ax$.

Reminder:
$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.
$f'(x) = \frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

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Exercise 12B – Pearson Pure Y1 page 261-262

12

Challenge\quad $\mathrm{f}(x) = \frac{1}{x}$.
a Given that $\mathrm{f}'(x) = \lim_{h\to 0} \frac{\mathrm{f}(x+h)-\mathrm{f}(x)}{h}$, show that $\mathrm{f}'(x) = \lim_{h\to 0} \frac{-1}{x^2+xh}$.
b Deduce that $\mathrm{f}'(x) = -\frac{1}{x^2}$.